

Camera & Telemetry

Monocular Images (30.3 Hz)
IMU / GT Telemetry
Intrinsic Matrix K

RAW VO Pipeline

1. KLT Feature Tracking
2. 5-Pt Essential RANSAC
3. recoverPose (SE3)

EIS Derotator Block

- SLERP Attitude Interpolation
 - Relative Rotation R_{rel}
- Homography $H = K R_{rel} K^{-1}$
 - Gating $|yaw_rate| > thresh$

EIS-GATED VO

Identical KLT & RANSAC
Evaluated on Derotated
Virtual Image Plane

(a) Un-derotated Baseline (b) Synthetic Derotation